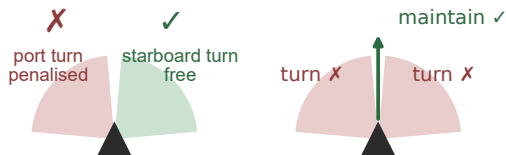


(a)

Directional non-compliance penalty

Head-On · Give-Way · Overtaking

Stand-On

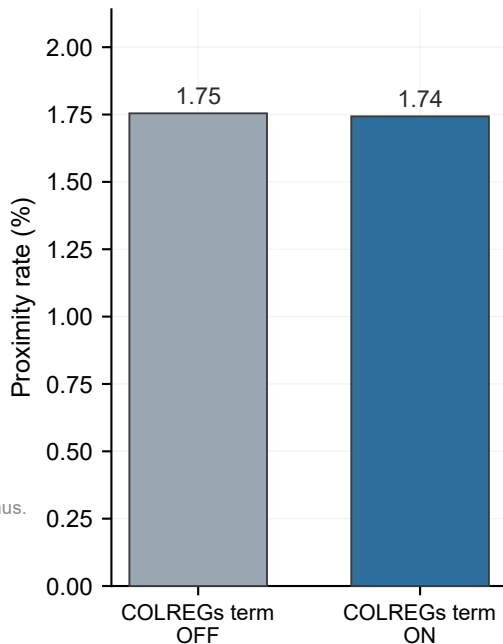


$$R_i^{\text{COLREGs}} = -\alpha_c (1 + \rho_i) \mathbf{1}[\rho_i > 0.3] (\dots)$$

A directional penalty on non-compliant turning — not a positive compliance bonus.
Weighted by collision risk ρ_i and applied only above a risk gate.

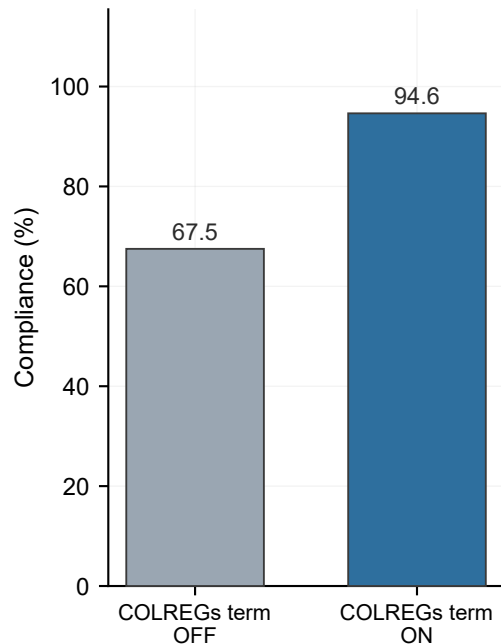
(b)

Proximity rate



(c)

Overall COLREGs (PRIMARY v2)



3 seeds. The COLREGs term raises PRIMARY v2 C (sit2/sit3 especially) and is the only ablation that moves C off the ceiling.
ON is residual+L2 hub; OFF is v11 freeze without the term. Arrival and proximity are now essentially tied; pick training by C here, not by proximity.